

Machine 4: A Sparse-Coincidence Cryptid with a Recovery Potential

The $A(a, b)$ system — halting on the line $b = a + 3$, and a no-cycle proof the plain recipe could not give

1. Overview

Machine 4 is the deterministic system $A(a, b)$ on positive-integer pairs, ten guarded rules (Section 2), started at $A(1, 1)$. It is an instance of the **sparse-coincidence** type: halting requires the orbit to land exactly on a line. Two things make it worth its own report. First, its no-cycle proof does *not* come from the collection's usual affine-potential recipe — no linear combination of a and b is monotone — but from a strictly weaker **potential-with-recovery** argument (Theorem 4), the first time the collection has needed it. Second, its growth is linear and its orbit comes startlingly close to halting (within distance 1 of the halt line), making it the collection's sharpest convergent cryptid.

Results: a stays odd forever, so the even- a rules never fire (Theorem 1); halting is exactly the line $b = a + 3$ with a odd (Theorem 2); the dominant rule cascades in closed form (Theorem 3); the machine provably cannot cycle (Theorem 4); and it did not halt in 3,000,000 steps, growing only linearly. The open core — whether the orbit ever lands on $b = a + 3$ — is the same single-orbit question as everywhere in this collection.

2. The machine

State $A(a, b)$. Dispatch: even $a = 2k$ uses the first three rules; odd $a = 2k+1$ uses the rest (on b relative to a).

rule	region
$A(2k, 1) \rightarrow A(3, 2k+2)$	even a , $b = 1$
$A(2k, 2) \rightarrow A(2k+1, 1)$	even a , $b = 2$
$A(2k, b) \rightarrow A(2k+3, b-2)$, $b \geq 3$	even a
$A(2k+1, 2m) \rightarrow A(4m+3, 2k-2m+1)$	odd a , $b = 2m \leq a-1$
$A(2k+1, 2m+1) \rightarrow A(4m+3, 2k-2m+3)$	odd a , $b = 2m+1 \leq a$
$A(2k+1, 2k+2) \rightarrow A(4k+1, 1)$	$b = a+1$
$A(2k+1, 2k+3) \rightarrow A(4k+7, 1)$	$b = a+2$
$A(2k+1, 2k+4) \rightarrow \mathbf{HALT}$	$b = a+3$
$A(2k+1, 2k+5) \rightarrow A(4k+5, 1)$	$b = a+4$
$A(2k+1, b) \rightarrow A(4k+7, b-2k-5)$, $b \geq 2k+6$	$b \geq a+5$

Fidelity. The implementation reproduces the hand-computed prefix $A(1,1) \rightarrow A(3,3) \rightarrow A(7,3) \rightarrow A(7,7) \rightarrow A(15,3) \rightarrow A(7,15) \rightarrow A(19,4) \rightarrow \dots$

3. Invariant and halting criterion

Theorem 1 (a is always odd). Every odd-a rule outputs an odd a (each right-hand side is $4m+3$, $4k+1$, $4k+5$ or $4k+7$), and $a_0 = 1$ is odd. So a is odd at every step and the even-a rules $A(2k, \cdot)$ are unreachable from $A(1, 1)$. (Verified: 300,000 random odd-a states and 200,000 orbit steps.)

Theorem 2 (halting criterion). With a odd, the only halting rule is $A(2k+1, 2k+4) \rightarrow \text{HALT}$, i.e.

the machine halts iff it reaches a state with a odd and $b = a + 3$.

(Verified exhaustively for $a < 4000$.) Since a is odd, $b = a + 3$ is even. This is the sparse-coincidence signature: halting is an exact landing on one line in the (a, b) plane. The four rules on the neighbouring offsets $b = a+1$, $a+2$, $a+4$ and $b \geq a+5$ all continue, threading around the halt line.

4. Acceleration and no cycles

4.1 Theorem 3: the dominant cascade

The rule for $b \geq a+5$, $(a, b) \rightarrow (2a+5, b-a-4)$, is iterable in closed form: after j rounds

$$a_j = 2^j(a+5) - 5, \quad b_j = b - (2^j - 1)(a+5) + j,$$

valid while $b_j \geq a_j + 5$. a doubles while b drops by about a each round, so a long run of the dominant rule collapses to one jump, with an interior landing on the halt line detected in closed form (20,000 runs verified against the base rule).

4.2 Theorem 4: no cycles, by a potential with recovery

Here machine 4 departs from its predecessors. No affine combination $p \cdot a + q \cdot b$ is monotone: the small-b rules $(a, b) \rightarrow (2b+3, a-b)$ and $(2b+1, a-b+3)$ shrink a sharply, and the $b = a+1$ rule lowers $a + b$ by 1. The collection's affine-potential recipe (pattern P6) simply fails. But $a + b$ is a **potential with recovery**:

Lemma A	$\Delta(a+b) \leq 0$ only for $b = a+1$ ($\Delta = -1$) and $b = a+4$ ($\Delta = 0$); both land on a state with $b = 1$.
Lemma B	from any $(a, 1)$ with a odd, $\Delta(a+b) = +4$.

Theorem 4. The machine cannot cycle. *Proof.* By Lemma A every non-increasing step is isolated (it lands on $b = 1$, and by Lemma B the next step increases $a + b$ by 4) and is immediately followed by a +4 step. Around any cycle the total change of $a + b$ is zero; but pairing each non-positive step with the following +4 gives a net $\geq +3$ for that pair, and every remaining step contributes $\geq +1$, so the total is strictly positive — a contradiction. Hence there are no cycles: the machine halts or escapes to infinity. **QED** (both lemmas verified on 400,000 random states and, for Lemma B, exhaustively to $a = 200,000$.)

This **extends the collection's no-cycle pattern**: the affine quantity need not decrease-free step by step, only over a bounded recovery window. It is worth recording as a reusable technique for the next

machine whose guards defeat a plain potential.

5. Behavior and status

The orbit grows **linearly**: $a + b$ drifts upward by about 2.34 per step (in 500,000 steps it decreased only once, by 1), reaching only ~23-bit values after 3,000,000 steps. So machine 4 is a **convergent-type** cryptid. It did not halt in 3,000,000 steps. Notably its closest approach to the halt line was $|b - (a+3)| = 1$ — it repeatedly reaches the immediately adjacent lines $b = a+2$ and $b = a+4$ (both of which continue), so the halt line is threaded rather than distant. Whether the orbit ever lands exactly on $b = a + 3$ is the open core; the branch that decides it reads ever-deeper digits of the state, the same single-orbit obstruction as the rest of the collection.

check	scope	result
Theorem 1 (a odd)	300,000 random + 200,000 orbit steps	pass
Theorem 2 (halt iff $b = a+3$)	exhaustive for $a < 4000$	pass
Theorem 3 (cascade)	closed form vs base rule, 20,000 runs	pass
Theorem 4 (no cycles)	Lemmas A, B on 400,000 states + $a \leq 200,000$	pass
deep run	3,000,000 steps, no halt, linear growth (~23 bits), closest $ b-(a+3) = 1$	pass

Artifacts: m4_base.py (the ten rules), m4_theorems.py (Theorems 1–4 and measurements). **Status: open (convergent, sparse-coincidence cryptid); halts or escapes — periodicity excluded by proof (via the recovery potential).**